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Breaking Free from Single-Use: Consumer Journeys and Perceptions in Packaging-Free Shopping

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ABSTRACT

Packaging-free grocery shopping enables consumers to avoid single-use packaging and buy specific quantities tailored to their needs. Despite its benefits, it faces challenges such as limited product variety and availability. This qualitative study explores the motivations, behaviors, and strategies of individuals who engage in or discontinue packaging-free shopping, including how they combine packed and unpacked products. Based on 115 open survey responses from packaging-free users across Europe, the study uses Activity Theory (AT) to analyze the interactions between consumers, food formats, and their goals within social and environmental contexts. The findings outline the consumer journey in combining packaging types, identifying tensions and harmonies within packaging systems. The analysis proposes a classification of consumer types based on behaviors, motivations and trip planning, offering insights for assessing environmental impacts, informing Life Cycle Assessments (LCA), and developing practical strategies for more sustainable grocery consumption.

KEYWORDS

Consumer behavior; packaging-free; Activity Theory; grocery; food packaging

Introduction

Packaging-free initiatives have emerged in response to the growing environmental concerns surrounding single-use packaging, particularly plastics, which have been highlighted as a significant contributor to global pollution (Geyer et al., 2017; Marazzi et al., 2020). This issue has prompted regulatory actions, such as the Single-Use Plastics Directive (EU 2019/904) implemented by the European Union. This directive aims to minimize the environmental impact of single-use plastics by limiting or forbidding specific products, consequently encouraging the use of reusable packaging systems (*European Parliament and of the Council*, 2019). Packaging-free systems offer consumers an alternative to single-use packaging by providing and encouraging the use of reusable containers. In grocery shopping, this often involves customers bringing their containers for refills or utilizing services where containers are provided, either in-store or through home delivery (Greenwood et al., 2021).

Packaging-free food shopping, as used in this paper, refers to grocery purchasing practices where products are obtained without conventional single-use consumer packaging. Typically, by refilling bulk dispensers or reusable containers instead of buying prepackaged items. In practice, this involves consumers bringing their own containers (e.g., jars, boxes, fabric bags) or using standardized reusable containers provided by the retailer, which are filled, weighed and labeled in-store or via refill and deposit-return systems. Packaging-free options are currently most common for dry foods such as grains, pasta, rice, pulses, nuts and seeds, as well as fresh produce, bakery items, coffee and tea, and

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some household and personal-care products (Ağlargöz & Ağlargöz, 2024), whereas highly perishable foods like meat and dairy remain more reliant on conventional packaging due to hygiene and shelf-life requirements.

However, while the concept of packaging-free systems was designed with environmental concern in mind, the practicalities of implementing packaging-free alternatives at a larger scale bring challenges. One major issue is the logistical complexity of offering packaging-free options for all product types, as many foods and household items are still primarily sold in conventional packaging formats (De Canio et al., 2024), making a complete shift difficult. Often, the conventional packaging system looks as it does for good reasons, as some products, especially perishable ones (Coelho et al., 2020) cannot easily transition to bulk formats or require specialized materials to maintain freshness, ensure safety, and minimize food waste (Verghese et al., 2015). Packaging-free options are never entirely free of packaging, as they always involve secondary and tertiary packaging. They therefore focus on eliminating only primary consumer packaging.

These constraints illustrate how packaging-free shopping is embedded within broader systems of food provision, where existing infrastructures, regulations, and material requirements shape the forms of consumption that are possible. Within these systems, packaging itself functions as a key artifact that enables storage, transport, safety, and convenience, while also constraining alternatives to conventional formats.

Moreover, while packaging-free alternatives reduce packaging waste, they may generate other environmental impacts. These include increased energy use associated with transportation or storage (Beitzen-Heineke et al., 2017) cleaning refill packages, unnecessary food waste or making additional trips to specialized stores where all needed items may not be available. Studies suggest that consumers who purchase loose products tend to face trade-offs between perceived affordability and concerns about hygiene or quality, which makes transitioning to packaging-free alternatives challenging for certain groups of consumers (Panda, Mukhopadhyay, et al., 2022).

In this study, such trade-offs are understood as tensions that arise when consumers navigate between different shopping systems and packaging in their everyday routines.

This implies that even if consumers are willing to adopt package-free shopping, they need, at best, to combine it with conventional shopping in their grocery routines. This combination can challenge the environmental benefits of packaging-free systems. Therefore, the reality of combining packaged and unpackaged products in everyday shopping may not always lead to the positive ecological outcomes these systems aim to achieve (Jiménez Romanillos et al., 2024). The impact of these hybrid shopping practices has been underexplored, and it remains unclear whether they truly lead to the positive ecological outcomes intended by packaging-free systems. The first step in assessing these outcomes is to gain a deeper understanding of the consumer journey and their decision-making in their hybrid shopping practices.

Existing research on packaging-free systems has mainly focused on their potential benefits, such as waste reduction and consumer awareness of environmental issues. For example, studies about their influence on brand, trust and loyalty (Barbosa et al., 2023). However, few studies have examined how consumers balance packaging-free with other grocery practices (Jiménez Romanillos et al., 2024). This study aims to explore how consumers manage this combination of packaging-free and conventional groceries. By examining how consumers engage with multiple systems and packaging, and how tensions shape their choices, this study aims to move beyond evaluating packaging-free initiatives in isolation.

More sustainable food production and consumption are needed to address pressing environmental challenges and support the transition to more sustainable systems. Packaging-free systems are trying to be part of this shift, and exploring the spectrum of behaviors around the packaging-free systems is crucial to determine if consumers are engaging in practices that can reduce the environmental impact. By focusing on these hybrid shopping routines, this research seeks to understand how packaging-free systems coexist with conventional approaches, rather than replacing them entirely. The two main research questions of this study are:

(1) How are consumers' grocery shopping habits and routines affected when combining packaging-free and packaged products?
(2) How do tensions within packaging-free grocery practices relate to different ways consumers engage with packaging-free shopping?

To explore these research questions, the study examines whether diet, location, and household factors influence consumer behaviors and their behavioral patterns. These factors could shape shopping habits, as they meet with lifestyle preferences, cultural norms, and accessibility to packaging-free options.

Activity Theory (AT) is used in this study as it offers a contextual way of examining how consumers interact with the tools, rules, infrastructures and social contexts that shape their grocery routines (Kaptelinin & Nardi, 2006). While approaches such as Theory of Planned Behaviour (TPB) or Social Practice Theory (SPT) are useful for understanding either individual motivations or shared practices, AT emphasizes the role of artifacts, conditions and contradictions within the activity system, making it especially suitable for activities where packaged and unpackaged systems co-exist. Drawing on Engeström's (1987) model, AT helps identify the tensions that arise between different activity systems, in this case between unpackaged and packaged shopping. This perspective highlights how consumers negotiate issues such as convenience, hygiene, store set-ups and personal values, and how these influence their choices.

This paper is organized into seven sections, beginning with a background on grocery systems and an introduction to Activity Theory. The methodology outlines the multistep coding process used to analyze the consumer practices. The results are presented in two parts: First, the Activity theory-based coding results, and second, the tension-based consumer grouping results. The paper concludes with a discussion and a conclusions section on how consumers choose between packed and unpacked grocery practices.

Background

Packaging-free grocery formats, including refill stations, bulk sections or reusable container options, have been introduced as alternatives to conventional single-use packaging, promoting new ways of performing grocery shopping (Fuentes et al., 2019). These formats require consumers to make different decisions and engage with products through weighing, refilling, and planning routines that are not present in packaged systems. Previous research has focused on environmental concern and personal responsibility, and sustainable values frequently motivate this shift (Marken & Hörisch, 2019). Meanwhile, other research has explored the operational aspects of packaging-free systems, from refill logistics to container management procedures in the industry and retail settings (De Temmerman et al., 2023).

Much of the research on packaging related to consumer behavior has been informed by intention-based models, particularly the Theory of Planned Behavior, which explains packaging choices through attitudes, perceived responsibility and normative expectations attached to sustainability oriented actions (Oliver et al., 2023). In parallel, social practice has also been applied, examining packaging-free and plastic-reducing consumption as sets of routines shaped by infrastructures, material arrangements and the competences required to perform them in everyday life (Rabiu & Jaeger-Erben, 2024). Together, these perspectives present the important motivational and structural dimensions of packaging-free consumption but often conceptualize packaging-free shopping as an individual activity rather than as part of everyday grocery routines and the coexistence of different packaging formats.

Previous research has also examined grocery shopping practices categories based on how consumers organize their shopping trips. Four main groups are identified based on store format preference and trip type: Planning Suburbans, who plan trips to larger supermarkets; Pedestrians, who conduct major shopping at nearby convenience stores to save time and effort; City Dwellers, who rely on frequent fill-in trips to convenience stores for accessibility; and Social Shoppers, who make fill-in

trips at supermarkets, often for social engagement (Nilsson et al., 2015). Previous research also highlights differing packaging choices between urban and rural consumers, showing that urban consumers prioritize informational attributes, such as expiry dates, and appreciate innovative designs that enhance convenience, whereas rural consumers focus more on practicality and familiarity in their purchasing decisions (Panda, Chakladar, et al., 2022).

Although current studies provide valuable insights into why consumers adopt packaging-free options and how practices are performed, they tend to examine packaging-free shopping as a disconnected activity rather than as part of a broader constellation of grocery routines (Jiménez Romanillos et al., 2024). As a result, unpackaged purchases are typically analyzed separately from the conventional shopping practices with which they are combined, limiting understanding of how packaging-free options are integrated into broader grocery decision-making.

Theoretical framework

The goal of this research is to identify the tensions that shape the routines of packaging-free and conventional grocery shopping. Understanding how consumers combine these systems will help assess whether grocery routines are environmentally sustainable and offer a foundation for designing practices that promote more sustainable behaviors. In the Design for Sustainable Behaviour field, theories such as the Theory of Planned Behavior (Ajzen, 1991), Social Practice Theory (Shove et al., 2012), and Activity Theory (Kaptelinin & Nardi, 2006) are commonly used to understand how behaviors develop (Selvefors et al., 2023). In this study, we apply AT, as it enables the analysis of behavior through the interaction between subjects, their goals, and the artifacts and systems they use, making it well-suited for comparing packaged and unpackaged grocery systems and understanding the role of mediating tools such as reusable containers. TPB explains behavior through individual intentions shaped by attitudes, norms, and perceived control, while SPT focuses on routines shaped by materials, competences, and meanings. However, the hybrid practices observed here involve system interactions, contextual constraints, and artifact-mediated actions that exceed the scope of intention-based or practice-based lenses. AT is chosen for its ability to reveal the system-level tensions and harmonies that structure consumers' grocery decisions. Table 1 compares AT with TPB and SPT to clarify the differences between lenses in the context of this study.

Activity Theory, originally developed by Vygotsky (1978) and later expanded by Leontiev Leont'ev (1978) and Engeström (1987), is employed in this study to extend the analysis of grocery shopping practices from a socio-cultural perspective. AT examines the interaction between individuals' goals, the artifacts they use, and the social structures that shape their actions, providing a comprehensive lens to study behavior (Leont'ev, 1978). This framework can offer insights into how environmental actions, such as reducing food waste or minimizing plastic use (Verghese et al., 2015), are influenced by personal motivations, the availability of artifacts, and the surrounding social context (Chu et al., 2021). By extending the model, Engeström introduced the concept of activity systems, which enables to examine contradictions and negotiations between different systems (Engeström, 1987). In the context of packaging-free shopping, the "system" includes the interconnected elements that make this practice

Table 1. Lenses comparation.

Theoretical lens	Primary unit of analyses	What drives behaviour change	Application in this study
Theory of Planned Behaviour	Individual	Changes in attitudes, norms, and perceived behavioural control leading to intention formation	Helps explain participants' environmental attitudes and perceived effort, but cannot capture system tensions
Social Practice Theory	Social practices	Reconfiguration of materials, competences, and meanings	Helps explain planning, storage, routines, and meanings of "eco-friendly" shopping
Activity Theory	Activity systems (subject, tools, rules, community)	How tensions emerge and are negotiated across interconnected activity systems	Captures packed vs unpacked system interactions, artefact mediation, and tensions central to this study

viable: reusable containers, refill stations, consumer habits, and the infrastructure that supports these activities. This approach provides a holistic overview of the packaging-free grocery system and how individuals navigate the balance between environmental motivations, practical challenges, and social influences (Kaptelinin & Nardi, 2006). To investigate these behaviors further, this study applies four key principles of AT: 1) the hierarchical structure of activity, 2) the expanded activity model, 3) the tensions within and between systems and 4) behavioral change and development. These principles guide the analysis of how environmentally conscious practices can be influenced by different shopping contexts.

The hierarchical structure of AT breaks down human activity into three levels: activities, actions, and operations. The activities are motivated by broader goals, such as reducing plastic waste or saving money. The actions are the steps consumers take to achieve these goals, like using reusable containers or shopping at stores with bulk formats. Operations are the behaviors within their routines that support these actions, such as consistently remembering to bring reusable bags. This structure, which can give us a systematic description of the consumer's doings, is used in this study to explain the differences between consumer groups' routines. It also shows how these motives influence whether packaging-free shopping becomes a lasting routine (Kaptelinin & Nardi, 2006; Leont'ev, 1978). Those three layers are shown in Figure 1.

Engeström's expanded model broadens this analysis by considering the social and cultural context. It includes the subject (the consumer, representing the individual or group whose actions and decisions are central to the activity), the object (e.g., reducing waste, which serves as the motivating goal or intended outcome driving the activity), and the artifacts (such as bulk stores, reusable containers, or digital platforms, which mediate and facilitate the subject's interaction with the object). It also includes the rules (social norms, expectations, and regulations that influence and constrain the behavior of the subject, such as societal attitudes toward waste reduction or packaging policies), the community (the social environment, including influences from friends, family or media platforms, which shape the subject's motivations and actions), and the division of labor (the distribution of responsibilities and roles within the activity system, such as the shared efforts in achieving waste reduction). Lastly, the outcome refers to how grocery shopping is practically performed when purchasing unpackaged products, resulting in different practices around unpackaged consumption (Engeström, 1987). Together, the expanded model can capture grocery shopping as a socially situated activity shaped by multiple elements.

Tensions in AT highlight the challenges within and between activity systems. These tensions, or contradictions, arise when elements of a system, such as artifacts, rules, or objectives, are in conflict or when neighboring systems interact. They can act as mechanisms for change. This study focuses on second-level tensions occurring within a system (e.g., between environmental motivations and limited packaging-free options) and quaternary tensions which emerge between systems (e.g., conflicts between packaging-free and conventional grocery systems) (Engeström, 1987). By examining these tensions, this study identifies both barriers to and drivers of adopting practices with less single-use

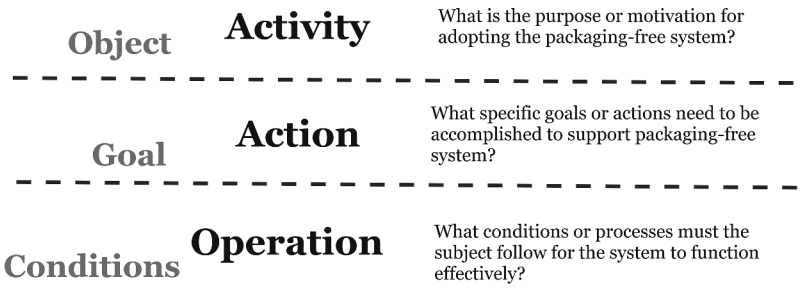


Figure 1. Hierarchical structure AT in packaging-free systems.

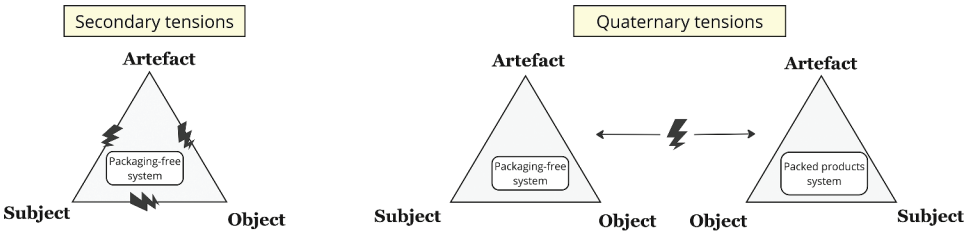


Figure 2. Secondary and quaternary tensions representation.

packaging. Secondary and quaternary tensions are represented in Figure 2 as they will be presented in the document.

In contrast, in this study, we will also refer to harmonies as the alignments within or between elements of an activity system that enable or support the goal (Selvefors et al., 2023). Harmonies can emerge when artifacts, rules, and objectives within a system are aligned or when neighboring systems collaborate effectively. For example, harmony might be observed when environmental motivations are supported by sufficient access to packaging-free options, or when packaging-free and conventional grocery systems complement rather than conflict with each other.

Studying behavioral change and development through AT provide an understanding of the shift toward new practices. Behavioral change involves individuals adapting their actions and habits, while the development of an activity refers to the evolution of systems in response to tensions and innovations (Kaptelinin & Bonnie, 2009). In the case of packaging-free grocery systems, behavioral changes can include consumers adapting to new shopping routines, while system development involves changes like expanding bulk store networks or offering packaging-free products within new types of systems. Through this framework, we seek to understand how consumers integrate packaging-free systems into their routines and how artifacts, rules, and norms evolve to support or hinder this transition, offering insights into the possible systemic changes.

Methodology

This study employs an open-answer survey to gather qualitative insights from consumers who have or are engaged in packaging-free grocery shopping. The survey was designed to collect data about their experiences, motivations, and challenges related to packaging-free shopping practices. Unlike quantitative surveys, this open-answer survey aimed to dig into the subjective experiences and personal insights of the respondents (Braun et al., 2021), collecting 115 answers, allowing for a deeper understanding of the factors that drive or challenge the adoption of packaging-free alternatives. This qualitative approach allowed participants to provide detailed and personalized responses, offering valuable insights into their experiences (Lim, 2024) with packaging-free shopping. A key advantage of this method was that it facilitated data collection across multiple languages. By analyzing the collected data, the study seeks to identify key behaviors and explore the diverse motivations and challenges consumers face in adopting packaging-free practices. This method was preferred over interviews or focus groups because the online open-ended format enabled diverse, context-rich responses without the geographical and scheduling constraints that would have restricted participation and reduced data variability.

The survey contained a series of open questions designed to explore aspects of the respondents' grocery shopping habits, from those consumers who tried to use packaging-free systems but stopped using them to those who included and continued using them in their routine. None of the questions were compulsory to answer, giving flexibility to the participants to not answer specific questions. The survey covered topics such as initial motivations for adopting packaging-free alternatives, the types of packaging-free methods and containers used, changes in grocery shopping routines, factors affecting

the decision to continue or discontinue packaging-free shopping, and the perceived advantages and disadvantages of this method. Additionally, demographic questions were included, such as the consumer's location (country, urban or rural), household type and dietary habits. An initial pilot survey (Sampson, 2004) was conducted to validate the question design and the quality of answers.

The survey remained open for 245 days, with an average response time of approximately 12 minutes per participant. The survey was closed after observing that data saturation had been reached since the final 15 responses provided no new insights relative to prior answers submitted.

The survey was structured with guidance from key constructs in Activity Theory, which provided a framework for organizing the types of information required to understand packaging-free grocery practices. AT distinguishes between the motive of an activity, the goal-directed actions through which that motive is pursued, and the routine operations shaped by contextual conditions. These distinctions informed the development of questions addressing why respondents engaged with packaging-free shopping, how they carried out their grocery routines, and which contextual factors influenced their behavior. In line with AT's emphasis on mediating artifacts, the survey also included questions about the tools and materials used during shopping, as well as the broader contextual influences shaping participants' practices.

Accordingly, the survey was designed to gather data on the artifacts consumers rely on, but also the subjects and objects involved in the activity, the development of practices over time, and the tensions and harmonies that emerge when combining packaging-free grocery shopping with other purchasing practices. This AT-guided structure ensured that the survey captured both elements of the activity system and the dynamics that influence how different grocery practices coexist.

Participants were recruited from across Europe to capture a wide variety of uses with different routines, diets, household compositions, and access to packaging-free options. Limiting the study to one city or country would have restricted this diversity and prevented from understanding those systems in different contexts, such as urban and rural variations. Participants were reached through several online platforms, including Facebook groups focused on zero-waste and plastic-free lifestyles, LinkedIn, university research networks, and posters displayed in grocery stores across multiple countries. The survey was available in Bulgarian, German, Spanish, French, Italian, Dutch, Portuguese, Swedish, and Turkish. Each translation was reviewed by native speakers, who also assisted in disseminating the survey.

A Python script was developed to translate all answers into English. A random sample of translations from various languages was analyzed by native speakers to ensure accuracy and consistency in the translation process. The analysis involved coding the answers for each question to identify common categories and categorize responses accordingly. This process was carried out manually using multiple Excel sheets. For example, from the question, what or who motivated you to use packaging-free alternatives for the first time? The answers were categorized into the categories "environmental concern," "influence of family and friends," "influence of media and education," "practical reasons," and "past practices and traditions." Following the initial coding phase, we proofread the codes and classifications for accuracy and consistency.

A second round of coding was conducted to uncover patterns, draw connections among responses, and identify common denominators across groups. This second coding phase explores the potential commonality among users with different diets, households, and locations as commonalities in motivations, methods, and trip frequency. The consumer journey mapping methodology (Towers & Towers, 2021) was used in this phase, aiming to understand the sequence of decisions consumers make when engaging with packaging-free alternatives. Survey data were categorized into three stages: planning and trip, at stores, and at home, focusing on decisions such as motivations, trip planning, grocery choices, or waste management. Consumer groups were identified by thematic grouping of responses based on shared patterns.

Figure 3 illustrates the workflow employed to draw insights from the survey data.

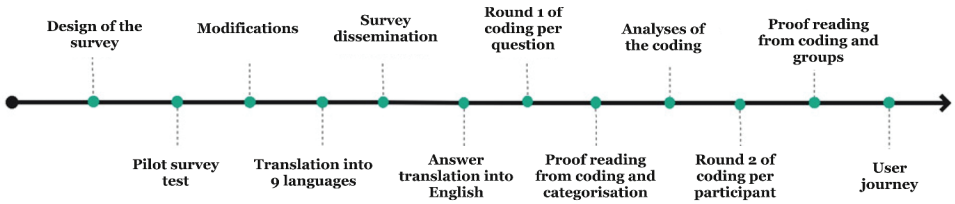


Figure 3. Methodological process.

The analysis was structured in five phases, following the principles of Activity Theory previously mentioned. The first four phases were analyzed from the insights of the first coding round. Initially, the analysis focused on some elements within the packaging-free system, such as artifacts (e.g., reusable containers) or objects (e.g., waste reduction), and the tensions and harmonies within this system. Following this, the second phase explored behavioral change and development within the packaging-free system, examining how consumers adapted their routines to incorporate packaging-free practices and how these changes reflected the systemic evolutions. The third phase analyzed interactions between packaging-free and conventional grocery systems, identifying key tensions of combining both approaches in daily routines. Lastly, the different categories from the questions were summarized into the extended AT triangle to represent the relationship between multiple elements and insights.

The fourth phase was analyzed using the insights from the second coding round. In this phase, AT's hierarchical structure was applied to participants' responses to examine their activities, actions, and operations. Based on their motivations, behaviors, and the conditions shaping their packaging-free practices, four distinct consumer groups were identified.

Overview of consumer demographics

Among the respondents, 81% (92 participants) reported continuing with packaging-free shopping routines, while 19% (21 participants) indicated they no longer maintain such practices.

Geographically, respondents predominantly reside in Spain, Portugal, Sweden, and Germany. The majority, 96 individuals, live in urban areas, while 10 participants are based in rural settings. In terms of dietary habits, 65% (75 respondents) reported having no specific diet, while others identified as vegetarians (16 respondents) and pescatarians (6 respondents). The remaining respondents cited various other dietary preferences. Many respondents came from single-person households, followed by couples without children, households with children, and shared accommodations. It is important to note that this study was entirely qualitative, and the figures presented above are descriptive rather than statistically representative.

Results

The results are presented in two parts. First, the Activity Theory coding results are structured around key elements of Activity Theory. Second, the tension-based consumer grouping and the corresponding consumer journey patterns are presented. For a description of the sample characteristics, see the "Overview of Consumer Demographics" section above.

Activity theory coding results

Object and outcome

The collective reduction of packaging waste and its associated environmental impacts, as shown in the insights from the survey, defines the object of packaging-free grocery shopping together with cost-saving, perception of healthier and local purchases, and following community traditions. This object represents both individual and societal commitments to addressing environmental challenges, linking personal actions to broader ecological goals. The outcome of this activity extends beyond simply reducing waste but also promotes mindful consumption, healthier lifestyle choices, and the integration of environmental practices that balance innovation with tradition.

The main categories that emerge from participants' responses to practice packaging-free shopping include environmental awareness, practical and economic considerations, health and well-being, and social and cultural factors. Environmental awareness was the most prominent motivation, reflecting participants' alignment with environmental efforts. Statements such as "*curiosity and concern for the environment*" and "*the drive to reduce the amount of packaging waste I generate*" highlight the connection between individual actions and collective environmental responsibility.

Social and cultural factors also significantly influenced participants. Friends, family, and online communities emerged as important drivers, with respondents citing "*a friend*" or exposure to "*Instagram vegan and sustainable food influencers*" as motivators. Family traditions further supported engagement; for instance, one participant explained, "*My mum used bags for vegetables and gave me such bags*," presenting intergenerational habits that encouraged those practices.

Practical and economic considerations were highly mentioned as motivators as well. Many participants appreciated the ability to control quantities and reduce food waste, with one respondent highlighting the convenience of buying "*the amount of food I actually needed to avoid food waste*." Others noted the cost-effectiveness of packaging-free shopping, especially through bulk purchases that reduced overall spending. Economic motivations extended to avoiding unnecessary packaging expenses, such as "*the fact of not using plastic bags*" or packaging that "*adds no value (e.g., toothpaste in a box to look better on the shelf).*"

Lastly, health and well-being were important to some participants. They associated packaging-free products with being *"fresher and having fewer preservatives,"* aligning with their desire to "*live healthier*."

Figure 4 shows the main objects and connected outcomes of packaging-free systems.

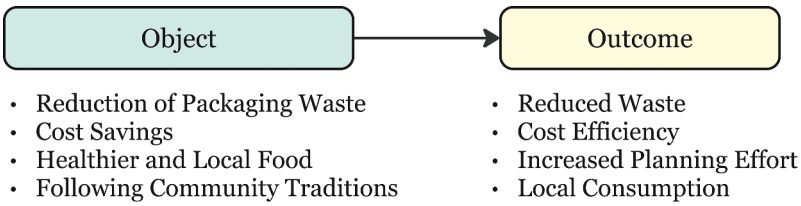


Figure 4. Object and outcome in the packaging-free system.

System and Artifacts

Types of Packaging-Free Systems Used

Participants used a variety of packaging-free systems, such as refill stations, online delivery services, farmers' markets, and supermarkets offering loose products.

Among these systems, refill stations were frequently mentioned by the participants as a favored alternative. These stations enable consumers to refill their containers with products such as nuts, grains, and household liquids. A participant who regularly uses refill stations for nuts and some dry products noted that "*it's not easy to find inside supermarkets*," highlighting the preference for unpackaged products but the inaccessibility in supermarkets. Some respondents also choose online delivery and subscription services for their convenience. Particularly when these services offer packaging-free or minimal packaging options. However, participants do not rely solely on online services. They combine them with other systems, as listed by one respondent: "*Online delivery, own reusable bags for veggies and fruits, refill stations*."

Local markets and farmers' markets are often preferred for purchasing fresh produce, offering a way to avoid excessive packaging while supporting local economies. Many participants mentioned shopping at these markets for loose products and the option to bring their own bags. One respondent stated, *"More available without plastic at the market. I bring my own containers to the supermarket,"* stressing the adaptability of packaging-free practices across different shopping venues. Supermarkets are considered a practical choice for many participants. They often seek out supermarkets that offer loose produce or have specific "no packaging" policies for certain items. However, most participants agree they could not find all their preferences in conventional supermarkets. One respondent reflected on their strategy to deal with this: "*Bulk purchasing in the market, combining the trip for food products to the supermarkets*," again expressing the need to integrate multiple shopping practices to achieve their consumption goals.

Tensions and harmonies in packaging-free systems. Packaging-free shopping, while initiated to reduce waste and promote environmentally conscious grocery alternatives, comes with a set of challenges and tensions that complicate its adoption. One of the most significant difficulties is the additional time and effort required. Many find it "*more time-consuming*" to shop and organize items at home. Preparing for such shopping trips involves remembering to bring containers, a seemingly simple task that, when forgotten, disrupts plans and the trip. This requirement for pre-planning removes the spontaneity from shopping, as one respondent noted, "*with unpacked products, it was impossible to pass by the stores unplanned, as you always need to plan*."

Another common tension arises from practical and logistical challenges. Carrying reusable jars, bottles, or other packaging materials is often inconvenient and, for some, "*not economically and practically worth it*." Storage and freshness of unpackaged products present further difficulties, as items "*do not keep as well in the fridge*," leading some consumers to prefer prepackaged products that offer a "*stronger sense of cleanliness and freshness*."

Accessibility and availability of packaging-free stores are other critical issues. Many respondents expressed frustration over the limited number of such stores, with comments like "*packaging-free stores aren't so widely spread out*." The limited product selection in these stores adds to the difficulty, as consumers often cannot find all the items they need in one place, forcing them to turn to conventional supermarkets. While some find packaging-free shopping cost-efficient, others feel it is "*slightly less competitive in terms of price*" compared to conventional options. For many, economic constraints make it difficult to justify the added effort. Psychological barriers, such as a preference for prepackaged items that feel cleaner or more hygienic, further complicate the adoption of packaging-free practices. Those tensions between the Artefact (products and types of packaging), the subject (the four consumer groups in this case), and the object (the objective) are presented in Figure 5.

Despite the challenges, packaging-free shopping also brings some rewards and harmonies that appeal to users and non-users alike. One of the most frequently mentioned benefits is the opportunity to engage in environmentally sustainable practices. Many participants feel that "*not generating so*

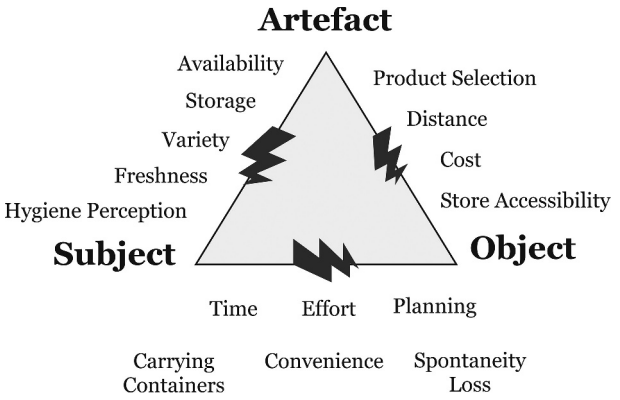


Figure 5. Tensions in packaging-free systems between the subject, the artefact and the object.

much garbage is satisfying," as it aligns with their environmental values and contributes to reducing waste.

Cost efficiency also contributes to these harmonies, particularly for those who take advantage of bulk buying. Shoppers find it "*cheaper to buy in bulk*," which not only reduces waste but also simplifies storage at home.

A sense of community and shared values further strengthen the positive aspects of packaging-free shopping. Many participants feel they are part of something larger, a movement that promotes eco-friendly and collective action. As one participant noted, there is a "*more positive shopping experience – feeling that you are part of something good*." Social networks and community support within the zero-waste movement encourage participation and reinforce shared values, making it easier for individuals to feel connected and motivated to continue these practices.

Figure 6 shows the harmonies between the Artefact (products and types of packaging), the subject (the consumers), and the object (the objective).

Reusable packaging as artefact

Within packaging-free systems, reusable packaging emerges as a key artifact, reflecting respondents' adaptive and environmentally conscious habits. Respondents use several alternatives to single-use

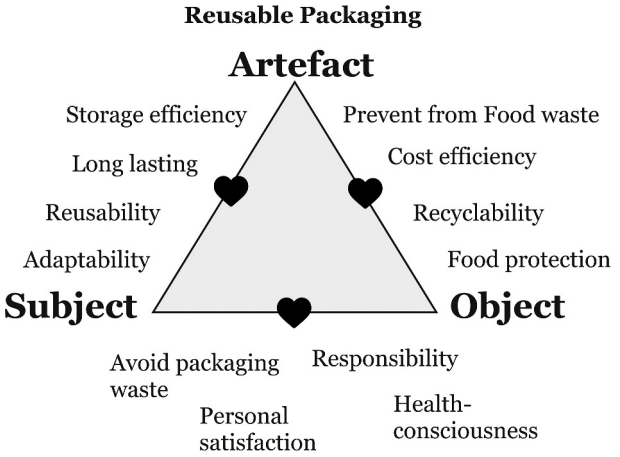


Figure 6. Harmonies in packaging-free systems between subject, artefact and object.

packaging, showing a strong preference for reusable bags combined with other packaging. These include cloth bags of all sizes and durable cotton and nylon shopping bags, valued for their environmental benefits and durability. Plastic bags and containers are also popular for their longevity and storage utility, with one respondent highlighting, "*I have been using one for 4 years, and it is still in good condition*." Meanwhile, glass containers as screw jars and bottles, are chosen not only for their durability but also for their safety and recyclability, as one participant pointed "*because of microplastics present in other bottles*."

Metal containers like stainless steel food boxes are preferred for specific needs, such as storing meat and fish hygienically: *"It is hygienic, does not retain smells, and washes well*," commented one participant. Other, less common choices were paper and cardboard packaging, like paper bags. Overall, the packaging choices presented in the responses are reusable packaging options driven by durability, environmental concerns, and practicality.

Behavioral change and development

From the perspective of Activity Theory, the adoption of packaging-free grocery shopping represents a shift in the activity system of grocery shopping, driven by the shared object of reducing packaging waste. This transition involves changes in artifacts (reusable containers, bulk stores), routines, and external factors such as product availability and cost. Behavioral changes reflect how individuals navigate tensions between sustainability, practicality and convenience.

Including packaging-free alternatives has changed planning, product choices and the effort involved. For some, shopping has become more meticulously planned. Respondents emphasized the need to "*remember to take containers from home*" and make thoughtful shopping lists. Buyers consider what to buy, where to shop, and, in some cases, a larger initial investment and product selection now depend on where unpackaged options are available. Many mentioned turning to local markets or traveling to packaging-free stores in other cities, with one noting, "*I plan better and go once a month to the next city to the packaging-free store*."

Adopting these habits often requires additional effort. Some respondents reported that buying unpackaged goods "*takes longer and is more expensive than plastic-packaged products*." However, a group described the transition as straightforward, with responses like, "*It was a very easy adaptation*," and "*I don't think it has changed much*."

Several factors influence the adoption of packaging-free shopping. Product availability was a recurring concern, with participants mentioning "*low availability of products in the area*" or noting that "*desired products are not available*." Such limitations often lead to compromises, particularly when bulk stores are far away or out of stock. The distance to shops offering packaging-free options complicates routines, with respondents highlighting the inconvenience of traveling to the "*store in the next big city*" for a broader selection.

Financial considerations also play a key role. While bulk purchasing can reduce costs for items like fruits and vegetables, participants noted that packaged goods are sometimes cheaper. As one respondent explained, "*If the packaged version is cheaper, I buy them packaged*." One participant remarked, "*If I'm out for another reason and want to shop, then I don't have containers with me*," emphasizing the need for advance preparation. Time constraints also affect decisions, as one stated, "*I need to plan in advance, and if I forget my bags and jars, I will buy them packed*."

This shift toward mixed shopping practices is illustrated in Figure 7 from the perspective of Activity Theory. Each system independently experiences its own set of tensions and harmonies, as shown in Figures 5 and 6. At the same time, these systems interact with each other, creating additional tensions between them, which are further explored later in this study and presented in Figure 8.

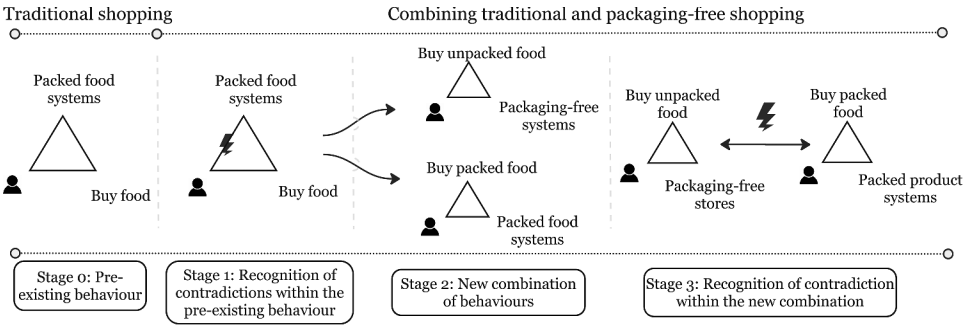


Figure 7. Stages in the development of the grocery shopping practice.

System combination & tensions

This section explores how individuals use a combination of the two systems and the challenges this creates.

Combination of grocery shopping systems

Participants reported various methods for grocery shopping, with distinct preferences shaped by convenience, time, or product availability. A common approach was to consolidate shopping into a single trip to a single store. This method appeals to those prioritizing efficiency, allowing them to complete all purchases at once. The primary motivation for this choice is saving time and reducing fuel consumption, as reflected in comments like *"I prefer to save trips and shop in one day."*

In contrast, other participants preferred shopping at multiple locations on different days to meet specific needs. However, this approach required more planning and flexibility, but enabled them to visit multiple packaging-free stores. They perceived these extra efforts as supporting more sustainable choices.

A hybrid method emerged among some participants, combining online and offline shopping. This approach maximized convenience while ensuring access to a broader range of products. Similarly, others alternated between packaging-free and conventional stores based on availability, choosing packaging-free options "*if possible*" but adapting to packaged alternatives when necessary. These findings highlight the adaptability of participants' shopping behaviors, driven by practicality and accessibility.

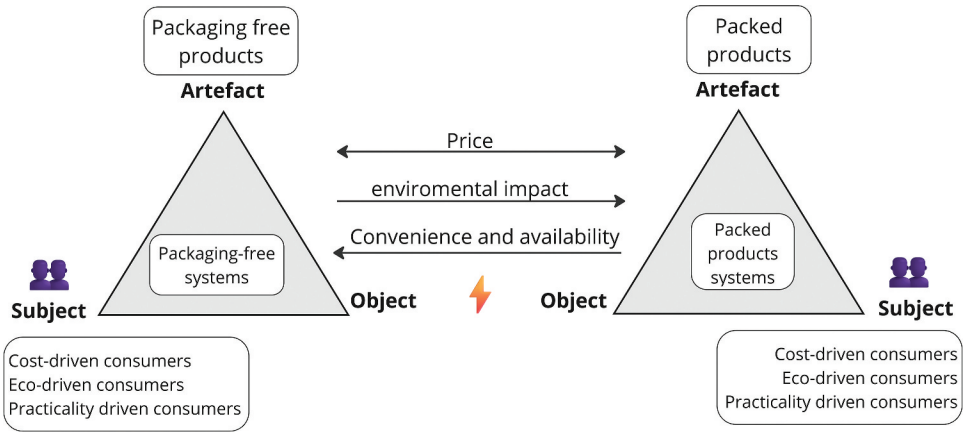


Figure 8. Main quaternary tensions between packaging-free and packaged product systems.

Tensions between packaged and unpackaged products

In practice, grocery systems are rarely used in isolation, most participants rely on a combination of systems, balancing packaged and unpackaged products. This blending of approaches emphasizes the tensions that arise when integrating environmental goals with the realities of grocery shopping.

When choosing between packaged and unpackaged products, participants faced multiple tensions. Price was a significant determinant, with many prioritizing the most affordable options due to financial constraints.

Environmental impact was another factor influencing decisions. Participants expressed a strong preference for unpackaged products to minimize waste and prioritize recyclability, also often considering the type of packaging material used.

Quality and health considerations also played a role. Participants often associated unpackaged products, particularly fresh foods, with higher quality. However, convenience sometimes tipped the scales in favor of packaged goods, especially for ease of transport, storage, or preservation. Availability and promotions led some to choose packaged items instead, as pointed out by a participant who "*usually opt for unpackaged ones, but I always check the label.*"

This interaction of factors reflects a balanced trade-off, aligning environmental motivations with practical realities like price, quality, and convenience. Figure 8 illustrates the main tensions between packaged and unpackaged grocery systems, presenting price, environmental impact, convenience, and availability as the key factors.

Tensions and harmonies for causing food waste

The impact of packaging-free shopping on food waste prompted mixed responses from participants, with many highlighting benefits such as better quantity control and increased awareness. One of the most significant advantages was the ability to purchase only the needed amount, reducing the risk of food waste. Participants described how this flexibility allowed them to calculate portions more precisely, avoid leftovers, and organize their fridges. For example, one participant explained that they "*can approximately calculate the portions per person*," while another participant indicated to "*buy what is only needed, thereby reducing the waste*." However, some participants noted difficulties in estimating the right quantities, leading to occasional over-purchasing. As one respondent acknowledged, "*Even if I can choose the quantities I want, I am still unsure of what amounts are right for me.*"

Participants reported that shopping for unpackaged goods made them plan meals more carefully and avoid unnecessary purchases. One participant noted becoming "*more aware*" of their consumption needs. This awareness often resulted in a visible reduction in food waste, with many emphasizing that they wasted less food overall by focusing on buying only what they needed.

Not all participants agreed that unpackaged shopping directly influenced their waste habits. Some argued that food waste was more dependent on individual planning and consumption practices than on whether the products were packaged or not. As one participant indicated, "*waste happens sometimes because items are left in the fridge for too long without being eaten*," suggesting that planning and usage habits remain relevant regardless of the packaging.

An additional tension was the perceived shorter shelf life of unpackaged goods, which sometimes necessitated faster consumption or alternative storage methods. Participants noted that unpackaged foods often required immediate use or freezing to avoid spoilage. As one participant explained, they eat unpackaged food "*right away or freeze it*" because of its shorter shelf life, while another highlighted that "*earlier food consumption*" becomes necessary when switching from plastic to glass storage.

Figure 9 presents the tensions between packaging-free and packaged products concerning food waste, highlighting quantity control, planning, and shelf life. It presents how packaging-free can reduce waste through flexibility, but also demands more planning and quicker consumption.

Tension-based consumer grouping results

Group of consumers and Behaviors

Through the analysis and coding of the answers, based on their motivations, tensions and behaviors, four consumer groups were identified. Each group present different priorities, whether driven by economic, environmental or practical considerations, which shaped their shopping practices, artifacts, and approaches to minimizing waste been noted in other studies on sustainable shopping behaviors (Kenny et al., 2023).

Of the 92 consumers who still continue packaging-free practices, 89 answers were categorized into groups, with 32 urban eco-driven consumers, 27 cost-driven consumers, 20 practicality-driven consumers, and 10 rural eco-driven consumers. The following sections describe each group and the tensions shaping their routines.

Cost-driven consumers

This group includes urban consumers who prioritize saving money. This group prioritize minimizing expenses, but packaging-free options, often seen as environmentally driven, are only embraced if they are cheaper. Economic motivations are similarly observed in studies on more sustainable food packaging choices, where cost-effective solutions are a priority for financially motivated consumers (Herrmann et al., 2022).

They usually stick to shopping at nearby supermarkets or local stores, often making spontaneous stops, with little advanced planning. As one participant put it, *"I buy everything on the same day and in the same store."* While they do make an effort to reduce food waste, its environmental impact is not the main reason. Some studies have shown that bulk purchases can make households save in the grocery, aligning with this consumer intention (Griffith et al., 2009).

Conditions such as the proximity of stores influence their behavior. They avoid extra travel and routine changes, relying on low-cost items, plastic containers and reusable bags.

The main tension arises when bulk or unpackaged goods are more expensive than packaged alternatives, pushing consumers toward cheaper options. Another tension is the limited product

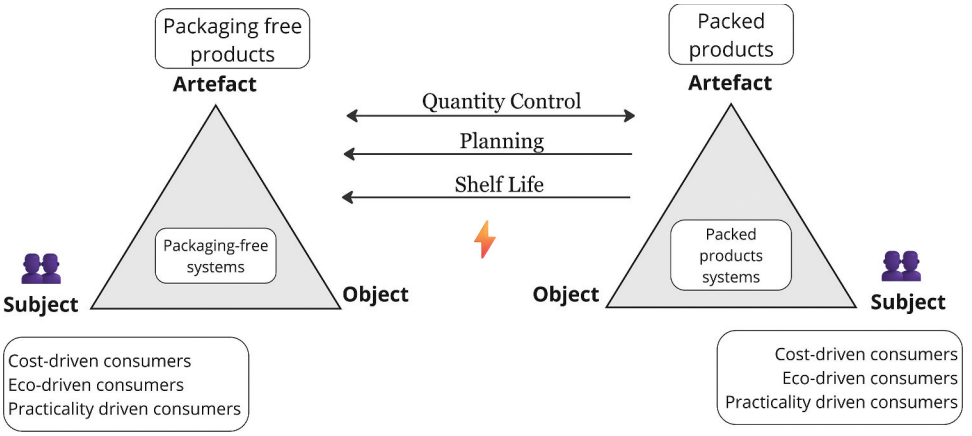


Figure 9. Main quaternary tensions between artefacts from packaged and packaged-free systems.

variety in packaging-free options. This group finds harmony when packaging-free options allow for cost-saving, such as in bulk buying. Harmony is stronger when bulk options both reduce cost and allow purchasing only what is needed, particularly when supermarkets offer these choices without disrupting routines.

Urban eco-driven consumers

This group consists of urban consumers prioritizing environmental impact reduction. Many participants emphasize their commitment to reducing plastic and other waste, viewing their shopping choices as a way to contribute positively to the environment. This commitment aligns with findings showing that urban consumers are increasingly aware of the environmental impacts of plastic and actively choose alternatives to minimize their waste (Heidbreder et al., 2019). For example, one participant said, "*I prioritise products without packaging or choose glass instead of plastic*."

Their goal is to minimize waste and shop in an environmentally conscious way, even if it requires more effort, time, or money. They carefully plan their shopping trips, visiting bulk markets, refill stations, and stores that offer packaging-free products. They are often willing to make extra trips, or travel longer distances to access packaging-free options. For instance, one participant noted, "*I don't mind travelling to bulk markets or refill stations because I feel like I'm helping the planet.*"

The conditions influencing their behavior include store availability distance and product variety. Their shopping habits involve using reusable bags, glass jars, and other eco-friendly containers to store their groceries, and they often buy in quantities that minimize waste.

In the eco-driven group, the most notable tension arises between convenience and commitment. Long travel distances and planning demands conflict with convenience. Despite these tensions, harmony is achieved when their environmental impact aligns with the artifact of packaging-free shopping, allowing them to actively reduce waste by using reusable containers and locally sourced food.

Rural eco-driven consumers

In rural areas, the same environmental motivation is shaped by different conditions. Packaging-free options are less accessible. Consumers often face long distances to reach stores with bulk format, creating a stronger tension between reducing environmental impact and the emissions or time associated with travel. Many participants also follow specific dietary practices, such as a vegan, vegetarian, or pescatarian diet, which further limits the available product range. One respondent noted the importance of "*availability of products, distance to the store,*" presenting the constraints they regularly encounter.

Despite these challenges, harmony arises when packaging-free options align with ecological values. Rural consumers frequently use jars, bags and glass containers to manage food storage and minimize waste, and they often plan purchases carefully to avoid over-buying.

Practicality-driven consumers

This group is motivated by the practicality of combining packaging-free grocery shopping with other activities in the urban context. While they are environmentally conscious, their primary focus is on convenience and efficiency. They appreciate packaging-free alternatives when these fit naturally into existing routines. One participant explained, "*An extra visit to some online stores, slightly lighter bags taken from regular stores*."

Their goal is to optimize their shopping process while reducing waste. They find packaging-free stores convenient when located along their commute or when paired with online orders. The conditions that influence this group's behavior include store proximity, online access and integration with other errands. For this group, the emphasis is on practicality in terms of time and convenience, although considering also cost savings, pointing out some common barriers to consumers regarding packaging-free products (Sang et al., 2022).

For the practicality-driven group, tensions arise from the conflict between time and environmental values. While this group support environmental practices, they prioritize efficiency and convenience. For example, forgetting containers or lacking nearby packaging-free options pushes them toward packaged goods. Harmony is achieved when the convenience of packaging-free shopping aligns with their daily routine, as well as other errands, online delivery and enables efficiency and reduced waste. If stores are located along their regular commute or if online delivery is available, they can streamline their shopping process while still reducing waste.

Consumer journey and system view

Drawing on consumer journey mapping as a conceptual tool (Shavitt & Barnes, 2020) this study applies it to identify patterns, motivations, and challenges that arise during the process. By visually representing these journeys, it becomes easier to compare certain characteristics of distinct consumer groups and gain a deeper understanding of how factors such as cost, convenience, and environmental awareness influence their decisions. This approach also highlights the trade-offs and tensions that different groups face navigating across the journey. As the map shows, environmental intentions are repeatedly negotiated against cost, convenience, and established routines at multiple stages. For example, when cost-driven consumers prioritize price over reusable packaging, while practicality-driven consumers rely on routine-based quantity choices rather than being determined by a single decision point.

Figure 10 represents the consumer journey as a linear process, providing clarity for comparing the behaviors of different consumer group choices. However, grocery shopping is a dynamic and repetitive activity, making a circular representation more appropriate. This format can also emphasize the circular profile of the reusable packaging present in the activity.

The circular journey map in Figure 11 also highlights the points of environmental impact and potential losses. Transport, for instance, involves decisions about travel distance, with longer trips increasing emissions, especially when private vehicles are used. Similarly, reliance on single-use packaging at the purchase stage contributes directly to waste generation or at the consumption stage, and food waste reflects the need for effective quantity management to minimize discarded perishables.

To complement this behavioral analysis, the extended AT triangle offers a systemic lens for situating individual behaviors within a broader context of societal and structural factors. The AT framework is expanded here to incorporate elements such as societal rules, community influences, and the division of labor, at a lower level mentioned by the participants of the survey, but still crucial to understanding the dynamics of the systems. Unlike the linearity of a journey map, the AT triangle functions as a synthetic representation that visualizes key elements that mediate consumer behaviors, including cultural norms that prioritize environmental responsibility, and the role of collective dynamics as family, friends, and societal expectations, as Figure 12 summarizes. Within this system, health and safety standards are experienced as rules that limit flexibility in container use and require additional preparation before shopping. At the same time, community members are perceived as motivating this practice, while different actors within the division of labor, as markets and retailers, influence how the practice can be carried out.

Discussion

This study contributes to consumption research by identifying different consumer patterns and their behaviors in combining packaging-free and conventional grocery practices with packaged products. Rather than framing packaging-free consumption as an outcome of individual intention, the analyses highlight how grocery practices evolve through interactions between motivations, material

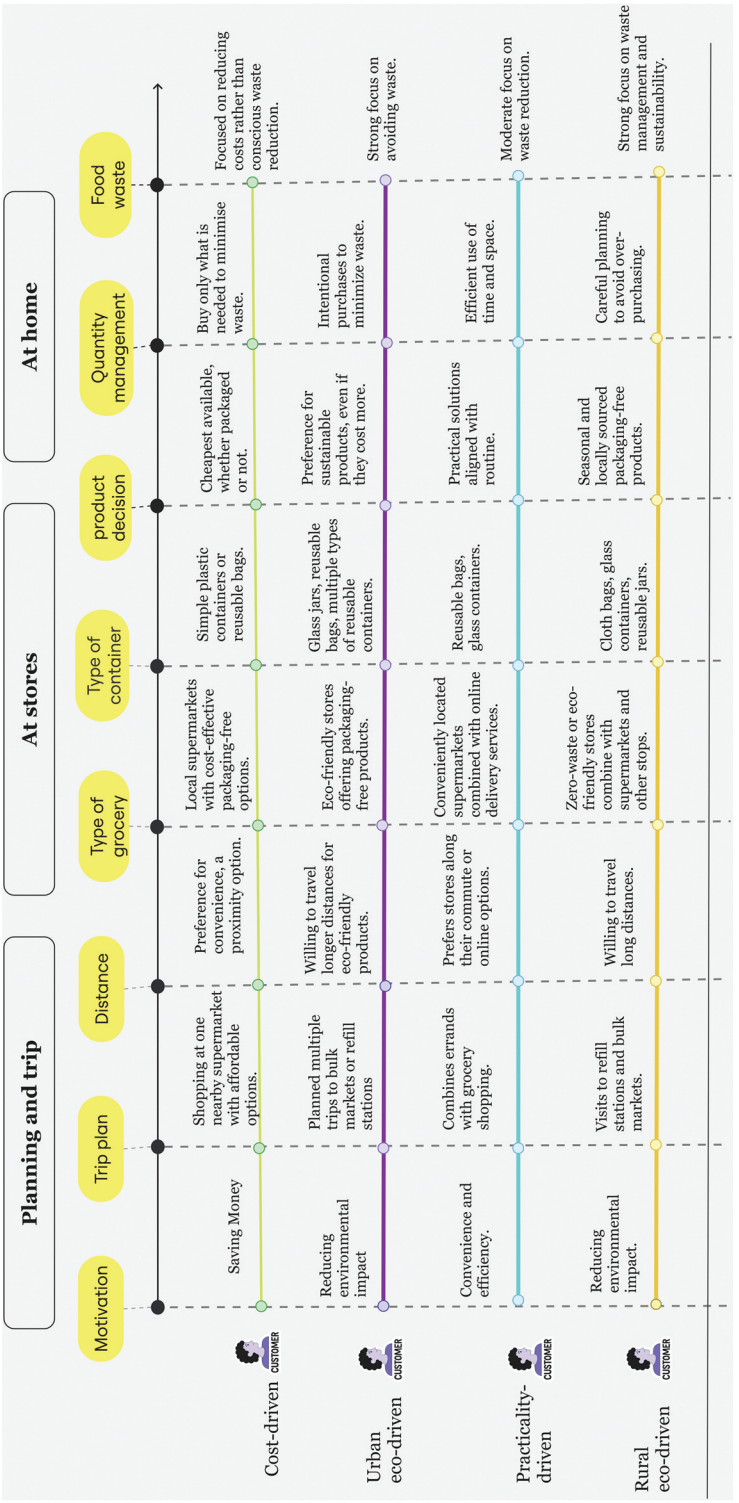


Figure 10. Consumer groups journey map.

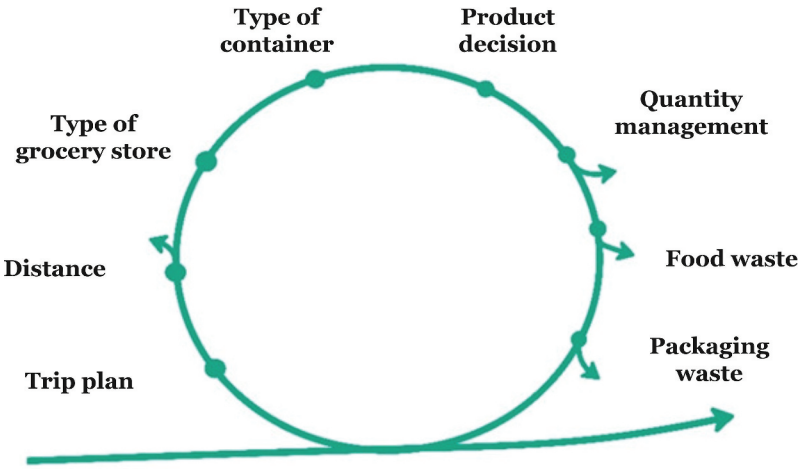


Figure 11. Circular journey map.

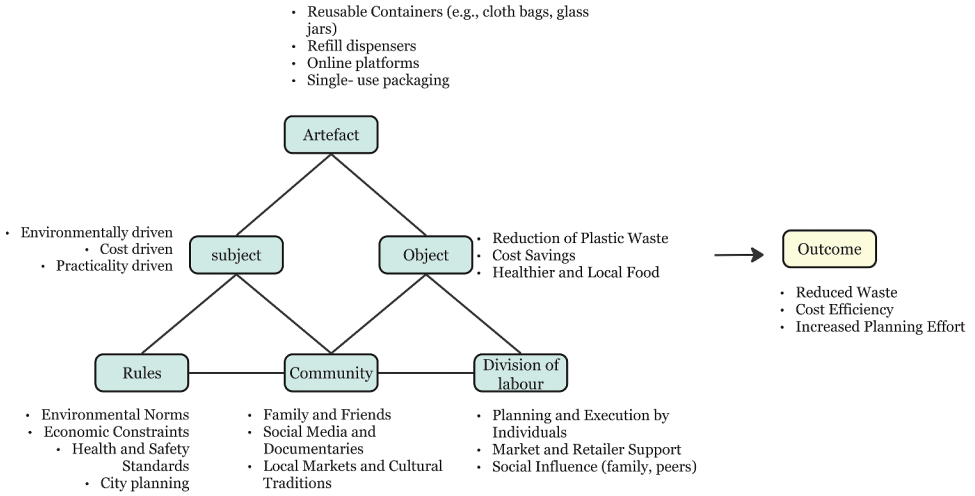


Figure 12. Packaging-free system overview.

arrangements, and infrastructure conditions. By applying Activity Theory, this research offers a structured approach to examining the dynamics of consumer routines, motivations, and contextual influences on their shopping behaviors. This systems perspective shifts attention away from consumer guilt or insufficient motivation toward structural limitations that restrict what even highly motivated consumers can do in practice. Furthermore, it provides valuable insights to assess whether the systems align with different environmental goals, enabling calculations on their environmental impact and supporting future calculations that incorporate behavioral patterns.

While four consumer groups emerged from the analyses, these groups are not discussed as fixed categories, but as ways of understanding how different tensions are negotiated within packaging-free grocery practices. Differences between economically motivated, environmentally motivated, and more practically oriented consumers reveal how cost, convenience, access, and effort shape everyday shopping routines when packaging-free options are introduced. Rather than motivation alone determining behavior, the findings suggest that consumers adapt their practices in response to systemic conditions, such as store location, pricing structures, and access to reusable packaging. In this sense,

the consumer groups help illustrate how packaging-free and conventional grocery systems are combined in practice, where tensions arise that either enable or constrain the sustained use of packaging-free options.

In this analysis, we engaged in identifying possible patterns related to diet, household type, and location, to uncover links to the different ways those consumer groups would behave during their grocery shopping. The limited influence of diet and household type suggests that packaging-free practices are shaped less by lifestyle characteristics and more by contextual and infrastructural conditions. Differences between rural and urban context therefore, reflect access and availability rather than fundamentally different environmental orientations.

While participants in this study demonstrated a clear internalization of environmental awareness and the need to reduce packaging waste, a gap between awareness and action emerged in some of the groups. Many participants engaged in packaging-free practices not to externalize their internalized ecological values but rather for reasons such as practicality, cost, or convenience. Linking to the AT concepts of internalization and externalization (Leont'ev, 1974), we see that while the problem of waste is well understood and internalized, it is not consistently externalized in behavior as an environmentally motivated action by practical and cost-driven consumers.

The findings show that the consumer journey of packaging-free shopping varies considerably between groups, with potential behaviors that can influence the environmental impact, such as trip frequency, distance, planning, storage practices, quantity management, or food waste. These variations show how attempts to reduce packaging waste can introduce new environmental trade-offs. Consumers who travel further to access packaging-free stores, for example, may offset their reduction in single-use plastics with higher carbon emissions, whereas those who prioritize one nearby supermarket can minimize transport impact but differ in the packaging waste and quantity management due to prepackaged food quantities. Additionally, the variety and origin of products, such as organically grown versus conventionally grown food, vary depending on the supermarket's type and size, influencing the overall environmental impact. This shows that packaging-free shopping does not automatically result in lower environmental impact when system configurations can steer consumers toward different behaviors. These variations stress the need for detailed life cycle assessments (LCA) to quantify the environmental impacts of each consumer journey.

However, it is important to note existing LCA studies on packaging-free systems, such as the work of Scharpenberg et al. (2021), often overlook the role of consumer behaviors in their analyses. This exclusion can lead to an incomplete understanding of the environmental impacts associated with varying consumer practices, such as travel frequency or waste management, which are crucial for a comprehensive assessment.

Conclusions

This study shows that shifting toward packaging-free grocery practices is not simply a matter of consumer motivation but depends on the alignment of systems, artifacts, and everyday routines. When environmentally preferable options remain inconvenient or costly, even highly motivated consumers struggle to sustain these practices over time. Packaging-free systems, therefore, require systemic support if they are to deliver meaningful environmental benefits.

Some limitations are the European-focused sample, drawn from consumers already engaging with packaging-free practices and consumers who stopped this practice. Furthermore, the study does not account for potentially influential variables such as income, which could further illuminate a narrower classification of the groups, particularly the cost-driven group. Lastly, the sample of consumers from rural areas was low, making it hard to find more than one consumer group.

When highlighting the tensions and harmonies within consumer practices, this study reveals how different choices meet particular challenges, leaving room for design interventions. While some consumers find harmonies between convenience and environmental goals, others encounter tensions, such as limited access to packaging-free stores or the added effort required to maintain the need for

reusable packaging systems. Understanding consumer behaviors can offer a foundation for designing practical and more environmentally conscious interventions. At a policy and business level, this points to the importance of artifacts that reduce consumer effort. For example, providing a larger variety of different sizes of packaged goods or introducing standardized reusable containers, expanding access to refill stations, or providing incentives for their establishment could help bridge the gap between consumer convenience and packaging-free accessibility. But before pushing any initiatives, future research should determine through LCAs whether and under which conditions these interventions promote more environmental grocery shopping alternatives and, if so, in which cases they do.

Author contributions

CRediT: **Elena Jiménez Romanillos:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing; **Suzan Boztepe:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing; **Helén Williams:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing; **Renee Wever:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Writing – original draft, Writing – review & editing.

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Data availability statement

The data collected in this study are available upon request from the corresponding author.

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